Types of Machine Learning

Module 3 Study Notes • Based on Amount of Supervision

1. Overview of Categorization

Machine Learning algorithms can be categorized based on various criteria. The most fundamental and widely used categorization is based on the **amount of supervision** the algorithm needs during its training phase. Based on this, ML is divided into four main types:

- Supervised Learning
- Unsupervised Learning
- Semi-Supervised Learning
- Reinforcement Learning

2. Supervised Machine Learning

Concept: Learning from labeled data. You provide the algorithm with both the input data and the correct output (the label or answer). The algorithm's job is to map the relationship between the inputs and the outputs so it can predict outputs for future, unseen inputs.

> **Analogy:** A student learning under a teacher's supervision. The teacher provides questions (input) along with the answers (output). The student learns the logic to solve new, similar questions.

Supervised learning is further divided based on the data type of the output (Target Column):

A. Regression

Occurs when the output variable is **numerical/continuous**.

- *Examples:* Predicting a student's final salary package (e.g., 4.5 LPA, 8.0 LPA), predicting house prices, or forecasting stock market values.

B. Classification

Occurs when the output variable is **categorical/discrete**.

- *Examples:* Predicting if a student will get placed ("Yes" or "No"), classifying an email as "Spam" or "Not Spam", identifying an image as a "Cat" or "Dog".

3. Unsupervised Machine Learning

Concept: Learning from unlabeled data. You provide the algorithm with input data only, without any corresponding outputs or labels. The algorithm's job is to independently find hidden structures, patterns, or groupings within the data.

Unsupervised learning encompasses four major techniques:

A. Clustering

Grouping similar data points together based on their inherent features.

- *Example:* Customer segmentation. An e-commerce platform grouping customers based on purchasing behavior to send targeted advertisements.

B. Dimensionality Reduction

Reducing the number of input variables (columns/features) while preserving the core information of the dataset. This solves issues related to slow processing times and the "curse of dimensionality."

- *Example:* Combining "Number of Rooms" and "Number of Washrooms" into a single feature like "Total Square Footage" (Feature Extraction like PCA - Principal Component Analysis). It is also heavily used to visualize 100-dimensional data by compressing it down to 2D or 3D scatter plots.

C. Anomaly Detection

Identifying rare items, events, or observations that deviate significantly from the majority of the data.

- *Examples:* Credit card fraud detection (flagging an unusually large transaction from a new location) or identifying manufacturing defects in a production line.

D. Association Rule Learning

Discovering interesting relations or rules between variables in large databases.

- *Example:* Market Basket Analysis. Discovering that customers who buy diapers are also highly likely to buy beer, prompting a supermarket to place them on adjacent shelves to boost sales.

4. Semi-Supervised Learning

Concept: A hybrid approach that uses a small amount of labeled data and a massive amount of unlabeled data. This is used because acquiring labels is often expensive, time-consuming, and requires human experts.

- *Example:* Google Photos. The system clusters thousands of your photos based on facial similarities (Unsupervised). It then asks you to label just one photo with a name (e.g., "Dad"). It automatically applies that label to the entire cluster (Supervised).

5. Reinforcement Learning

Concept: Learning through trial and error in an interactive environment. There is no historical dataset provided. An "Agent" takes actions within an "Environment" and receives either "Rewards" (for good actions) or "Punishments/Penalties" (for bad actions). The agent's goal is to maximize the cumulative reward over time.

- *Key Terms:* Agent, Environment, State, Action, Reward, Policy.
- *Examples:* Training self-driving cars, or AI agents that play complex games (like DeepMind's AlphaGo defeating world champions).